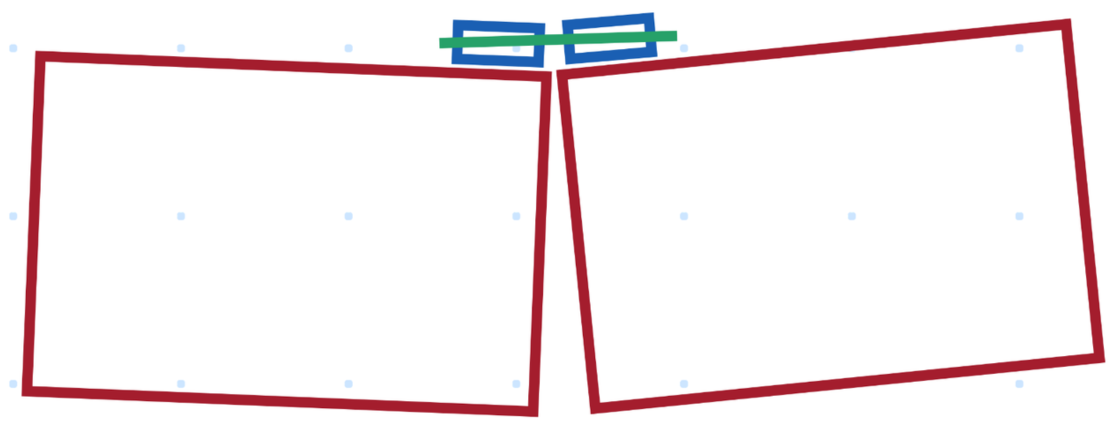
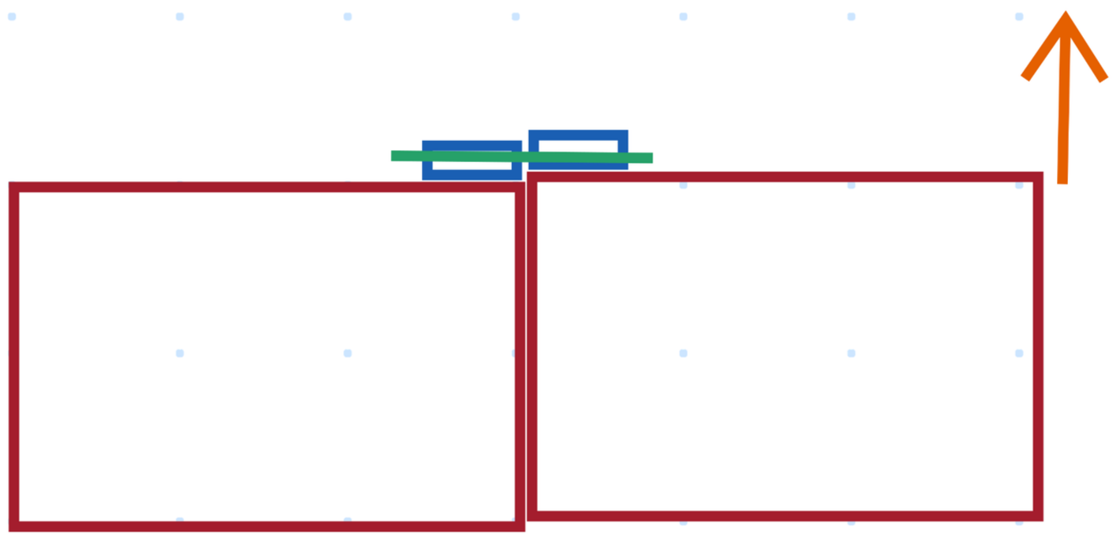
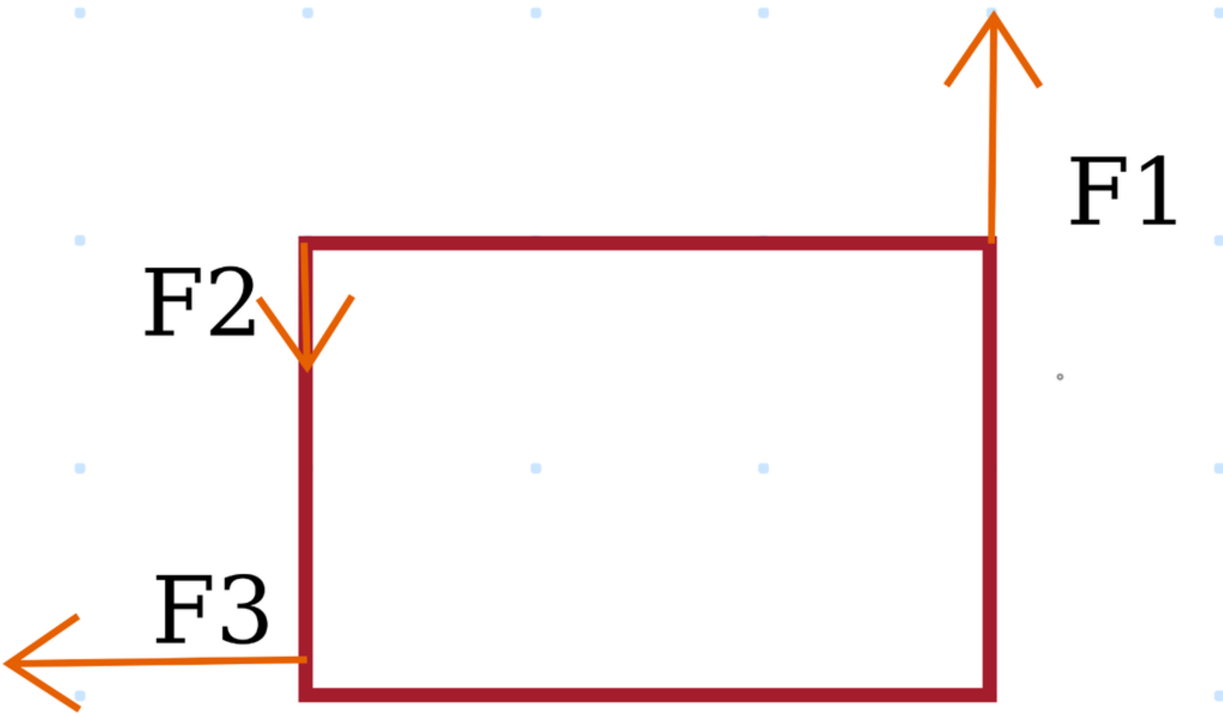
STEM principles guided our process of design refinement throughout the prototyping process, beginning with our considerations for the viability of the first prototype. With the calculations demonstrated in Element F, we determined the necessary cross-sectional area of steel to hold together the boxes with a deformation angle of 0.001 degrees (4 times the cross-sectional area of a single latch). With assistance from a research assistant at a nearby college with a bachelorette in mathematics and relevant engineering education, we refined our design to place the connection system at the top of the boxes as detailed in previous elements. We determined that the slack in the top connection system would induce a tendency to rotate in the boxes as shown in the below diagram:



Because of the difference in the outer diameter of the inner (green) cylinder and the inner diameter of the outer (blue) cylinder, the boxes are able to rotate while being picked up. Since only the boxes on the end will be held, this is likely to happen since the green cylinder will not support the middle box until it is able to touch the bottom of the rightmost box in the diagram (the one being lifted) and the top of the leftmost box in the diagram (the middle box in real life). In addition, since a force is applied to lift the boxes (shown in the following diagram) significantly away from the center of mass, the box would have a tendency to rotate until the torque produced by the interaction with the green cylinder and the blue cylinder opposed the torque produced by the person lifting the boxes and reduced to zero (this is possible since the net torque on the boxes as a system is 0 given that the boxes are being lifted from both ends). This would cause a considerable angle of deflection between the center box and the edge boxes as determined by qualitative analysis in a test of the connection system. In order to resolve this issue without requiring the inner cylinder to match the outer cylinder perfectly (which would make insertion difficult), we determined that a translational motion would be acceptable as shown in the below diagram because it wouldn't pinch the dragon so long as the translation was less than two times the thickness of the wall of the box:



We thus determined that we could place butterfly clips at the bottom of the side where the two boxes connect in order to oppose this torque. We described this with the following free body diagram:



From now on, we will refer to the forces labeled in the diagram as follows: F1 (the force exerted by the human) will be F_p, F2 (the force exerted by the cylindrical connection system) will be F_c, and F3 (the force rectifying the torque) will be F_r. We will assume the center of mass of the box to be at the center of the box for ease of calculations. We used the following calculations to determine the force on the butterfly clip as the box is lifted:

$$\sum \tau = 0 = \vec{r}_1 \times \vec{F}_p + \vec{r}_2 \times \vec{F}_c - \vec{r}_3 \times \vec{F}_r \text{ (note: } |\vec{r}_1| = |\vec{r}_2| = |\vec{r}_3| = r)$$

$$0 = rF_p cos(\theta) + rF_c cos(\theta) - rF_r sin(\theta)$$

$$0 = rF_p(\frac{w}{2r}) + rF_c(\frac{w}{2r}) - rF_r(\frac{h}{2r}) \text{ (note that h denotes height and w denotes width)}$$

$$0 = \frac{w(F_p+F_c)-F_r h}{2}$$

$$F_r = \frac{w(F_p+F_c)}{h}$$

We now determined the values of F_p and F_c to determine the value of F_r. Beginning with the rightmost box:

$$\sum F_y = ma_y = m(0) = 0 = F_p - F_c - F_g$$

$$0 = F_p - F_c - mg$$

Now solving for F_c using the same approach on the middle box:

$$\sum F_y = ma_y = m(0) = 0 = 2F_c - F_g \text{ (}F_c\text{ is same on both sides of the middle box)}$$

$$0 = 2F_c - mg$$

$$F_c = \frac{mg}{2}$$

Substituting this into the previously determined equation, we can solve for F_p:

$$0 = F_p - F_c - mg$$

$$0 = F_p - \frac{mg}{2} - mg$$

$$F_p = \frac{3mg}{2}$$

Now we can substitute these two values into the first equation to find the force on the butterfly clips:

$$F_r = \frac{w(F_p + F_c)}{h}$$

$$F_r = \frac{w(\frac{3mg}{2} + \frac{mg}{2})}{h}$$

$$F_r = \frac{2wmg}{h} = \frac{2(0.9144)(30.23)(9.81)}{0.508} \approx 1067N$$

Using this force, we can determine the stress on the butterfly clips and subsequently the number needed to ensure sufficiently small angles of deflection. The butterfly clips will be put under tensile stress since one side will be pulled towards the other box and the other side will be pulled towards the box on which the clip is attached, thus creating two forces that attempt to rip the clip apart. Using $\sigma = \frac{F}{A}$, we can determine the tensile stress:

$\sigma = \frac{F}{A} = \frac{F}{nA}$, where n is the number of clips and A is the cross-sectional area of one clip

$\sigma = \frac{1067}{(\frac{16}{(100)(10)})(\frac{1}{(100)(10)})(n)}$, since the clips cross-section is approx. 1 mm by 16 mm

$$\sigma = \frac{(1067)(10^3)^2}{16n} = \frac{(1067)(10^6)}{16n}$$

Using Young's modulus to determine the number of latches needed to keep the tensile strain under 0.001:

$$Y = \frac{\sigma}{\varepsilon}$$

$$Y = \frac{\sigma}{0.001}$$

$$Y = (\sigma)(10^3)$$

$$(200)(10^9) = (\frac{1067(10^6)}{16n})(10^3)$$

$$200 = \frac{1067}{16n}$$

$n = \frac{1067}{(16)(200)} \approx 0.3334$ Latches

Thus, using one latch per point of connection will ensure that the tensile strain remains under 0.001 in the latches, which is suitable for our needs. Since approximately 3 latches would be needed to keep the tensile strain around 0.0001, we can estimate that $0.0001 < \varepsilon < 0.001$ if we use two latches per point of connection, which is satisfiable for our needs. These calculations were confirmed by our engineering teacher, a licensed engineer in our state of residence and a well-established AP Physics 1 and C teacher. Through our iterative design process, we demonstrated the engineering portion of STEM principles, and through our mathematical modeling, we demonstrated the mathematical portion. Qualitatively, the inclusion of two butterfly clips was deemed necessary by the aforementioned research assistant to handle the torque generated by our discussed connection method. Furthermore, a civil engineer reviewed the connection system and determined it to be suitable for the komodo dragon in ensuring that the system remained stable under load.

This mathematical modeling sufficiently addresses the strength and durability design requirement while speaking to the reliability in terms of limited strain upon repeated use. For further analysis on the reliability criteria, we applied the aforementioned expert review to assume that the reliability would be sufficient for the success of the prototype prior to building it, but we also applied the scientific data and testing oriented principles to develop a method of testing for reliability as described in future elements. For safety, we furthermore developed methods of testing this, but we also included it throughout our calculations—the premise of the inclusion of butterfly clips was the safety of the enclosed animal. We determined that the rotation of the boxes could cause the pinching of the animal, which would lead to unsafe conditions. We applied the iterative design principles of engineering to refine our design while mathematically modeling it to ensure that the movement would be purely translational rather than rotational, which would ensure the safety of the animal since the likelihood of being pinched would be significantly reduced by sliding the boxes along each other with a very small separation rather than opening a wide gap in which the animal could become stuck.

In terms of ease of use, we again applied STEM principles to design around the wide margins of the upper connection system that were put in place to ensure the ease of assembly (since the pipes would be harder to slide together as the margins decreased). Addressing the material suitability, we chose to use steel for the butterfly clips because of its relatively high Young's Modulus compared to similar materials, which ensured a smaller strain for the same number of clips mathematically. We used a similar approach on the upper connection system to determine that steel would also be an ideal solution, but given the inability to find such a pipe, we used calculations in Element F along with the shear modulus of PVC (approx. 1 GPa) to determine that the cross-sectional area of the PVC needed to be approximately 0.00009358 meters squared to ensure a strain of less than 0.1 degrees. Since the cross-sectional area of 1-inch PVC is approximately 0.8647 square inches, or 0.0005578 meters squared, we determined that the angle of deflection would be less than 0.1 degrees for PVC, which is acceptable for our case given PVC's tendency to return to its original position after deforming. To ensure an angle of deflection smaller than 0.001 degrees (the number we choose in Element F), we determined that we would likely need to reinforce the pvc with steel, which we used to reform our design, thus demonstrating the application of engineering and mathematical principles. Further information on the calculations for this connection system can be found in Element F, and the aforementioned numbers were determined using a very similar method (solving for A instead of n). The calculations in Element F were verified by our engineering teacher.

We determined feasibility using the previous calculations along with calculations in Element F to ensure the feasible stability of the design while using the approval of the previously mentioned research assistant, engineering teacher, and civil engineer as validations of the feasibility of our design. In terms of cost-feasibility, we determined that the cost of the entire design would only be the cost of the butterfly clips (4) and the connection between the PVC and the box since the PVC and steel reinforcement were available as scrap in our engineering class. The total cost of the design was approximately $40, which falls well within our estimated budget (approximately $400 for the entire project, which would give us about $130 assuming an equal distribution of money).